



MEHRAN UNIVERSITY OF ENGINEERING AND TECHNOLOGY, JAMSHORO
DEPARTMENT OF ELECTRONIC ENGINEERING
INTEGRATED ELECTRONICS # 01



Introduction to Digital Integrated Circuits & perform logic functions using Multisim.

Roll No:	Date:
Checked by:	Score:

LAB PERFORMANCE INDICATOR	SUBJECT KNOWLEDGE	DATA ANALYSIS AND INTERPRETATION	ABILITY TO CONDUCT EXPERIMENT	PRESENTATION	DESIGN AND CODING	OBSERVATION/ RESULTS	SCORE

Performance Objective: Introduction of Digital Integrated Circuits & perform logic functions using Multisim

Apparatus:

- Digital Integrated Circuit IC's (74LS04, 74LS08, 74LS02, 74LS83 and 74LS85)
- Switches (SPST)
- Logic Analyzer
- Word Generator

Theory:

Digital circuits are circuits dealing with signals restricted to the extreme limits of zero and some full amount. This stands in contrast to analog circuits, in which signals are free to vary continuously between the limits imposed by power supply voltage and circuit resistances. These circuits find use in "true/false" logical operations and digital computation.

This technology came into being after the creation of transistor. Transistor revealed a dynamic change in this particular field. A monolithic integrated circuit (IC) is an electronic circuit that is constructed entirely on a single chip of silicon. All the components that make up the circuit-transistors, diodes, resistors and capacitors are an integral part of the chip. Integrated circuits providing a multitude of pre-engineered functions are available at very low cost, benefitting students, hobbyists and professional circuit designers alike. Most integrated circuits provide the

same functionality as "discrete" semiconductor circuits at higher levels of reliability and at a fraction of the cost. Figure-1 shows Dual-In-line Packaging of the IC.

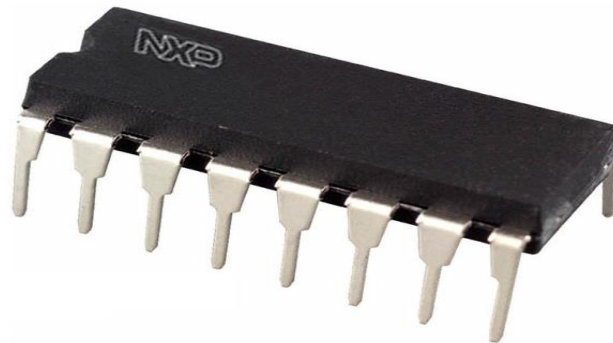


Figure-1 Dual-In-line Packaging of Integrated Circuit

Logic Families

Integrated circuits are design using the following logic families.

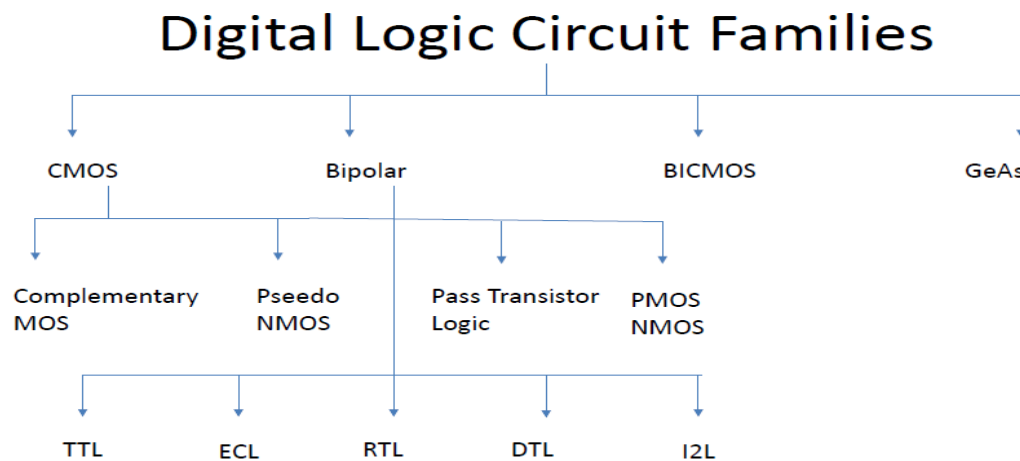


Figure-2 Logic families

Most of the digital Integrated circuits are design using either bipolar technology or using CMOS technology. The comparison between CMOS and TTL is given below.

CMOS LOGIC:

- **Dissipates low power:** The power dissipation is dependent on the power supply voltage, frequency, output load, and input rise time. At 1 MHz and 50 pF load, the power dissipation is typically 10 nW per gate.
- **Short propagation delays:** Depending on the power supply, the propagation delays are usually around 25 nS to 50 nS.

- **Rise and fall times are controlled:** The rise and falls are usually ramps instead of step functions, and they are 20 - 40% longer than the propagation delays.
- **Noise immunity** approaches 50% or 45% of the full logic swing.
- **Levels of the logic signal** will be essentially equal to the power supplied since the input impedance is so high.
- **Voltage levels** range from 0 to VDD where VDD is the supply voltage. A low level is anywhere between 0 and 1/3 VDD while a high level is between 2/3 VDD and VDD.

TTL Logic:

- **Power dissipation** is usually 10 mW per gate.
- **Propagation delays** are 10 nS when driving a 15 pF/400 ohm load.
- **Voltage levels** range from 0 to Vcc where Vcc is typically 4.75V - 5.25V. Voltage range 0V - 0.8V creates logic level 0. Voltage range 2V - Vcc creates logic level 1.

CMOS compared to TTL:

- CMOS components are typically more expensive than TTL equivalents. However, CMOS technology is usually less expensive on a system level due to CMOS chips being smaller and requiring less regulation.
- CMOS circuits do not draw as much power as TTL circuits while at rest. However, CMOS power consumption increases faster with higher clock speeds than TTL does. Lower current draw requires less power supply distribution, therefore causing a simpler and cheaper design.
- Due to longer rise and fall times, the transmission of digital signals becomes simpler and less expensive with CMOS chips.
- CMOS components are more susceptible to damage from electrostatic discharge than TTL components.

IC Types

In the early days of integrated circuits, only a few transistors could be placed on a chip. As the degree of integration was small, the design was done easily. Over time, millions, and today billions, of transistors could be placed on one chip, and to make a good design became a task to be planned thoroughly. This gave rise to new design methods. Integrated circuits are often classified by the number of transistors and other electronic components they contain:

- **SSI (small-scale integration):** Up to 100 electronic components per chip
- **MSI (medium-scale integration):** From 100 to 3,000 electronic components per chip
- **LSI (large-scale integration):** From 3,000 to 100,000 electronic components per chip
- **VLSI (very large-scale integration):** From 100,000 to 1,000,000 electronic components per chip
- **ULSI (ultra large-scale integration):** More than 1 million electronic components per chip

Classifications

Integrated Circuits can be classified into three types:

1. **Analog**
 2. **Digital**
 3. **Mixed Signals**
1. **Analog Integrated Circuits:** Analog ICs, such as sensors, power management circuits, and operational amplifiers, work by processing continuous signals. They perform functions like amplification, active filtering, de modulation, mixing, etc
 2. **Digital Integrated Circuit:** Digital integrated circuits can contain anything from one to millions of logic gates, flip-flops, multiplexers, and other circuits in a few square millimeters. The small size of these circuits allows high speed, low power dissipation, and reduced manufacturing cost compared with board-level integration. These digital ICs, typically microprocessors, DSPs, and micro controllers, work using binary mathematics to process "one" and "zero" signals.
 3. **Mixed Integrated Circuits:** ICs can also combine analog and digital circuits on a single chip to create functions such as Ac/Dc converters and Dc/Ac converters. Such circuits offer smaller size and lower cost.

Inverter logic using Multisim

In following figure shows how to connect word generator and logic analyzer with 74LS04 IC.

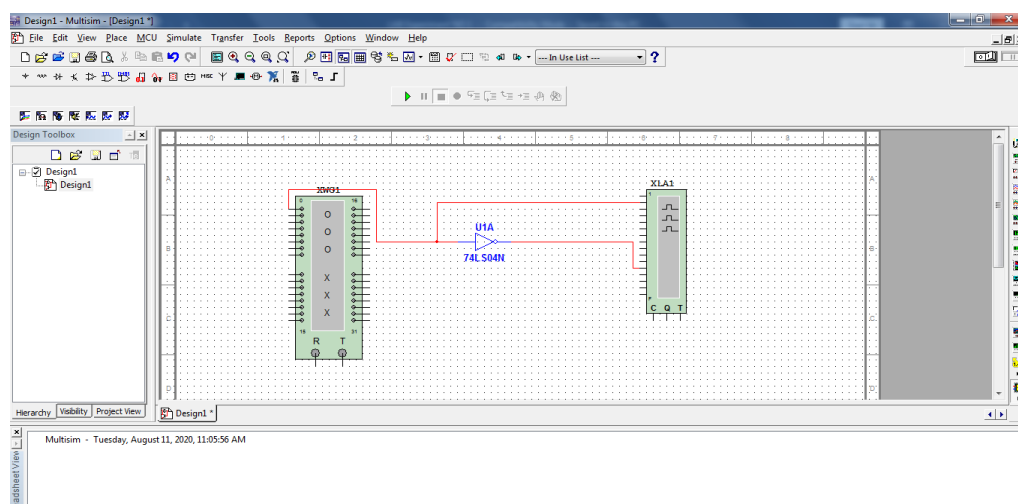


Figure represents how to connect inverter with word generator and logic analyzer

To apply the input sequence in word generator first double click word generator block, click setting apply desired sequence like binary, hexadecimal, octal etc and then set the desired buffer size, after it run the simulation and check your input and output waveform in logic analyzer.

Review Questions

1. Describe in detail the TTL, RTL, ECL and DTL Logic families(Attach extra sheet if required)

TRANSISTOR TRANSISTOR LOGIC= family is the fastest logic family ever and one of most widely used logic family for small and medium scale devices used for VLSI .These have capability of becoming part of complex devices into simpler devices .

RESISTOR TRANSISTOR LOGICfamily it replace the device switch with the

Transistor switch give efficiency The NOR gate of RTL is the basic .Resistor resistor logic has poor noise

margin and poor fan out capability and has high power dissipations .

EMITTER COUPLED LOGIC= it is fastest circuit family it also available for

conventional logic system design .high speed achieved by operating all bipolar transistor out of saturation.

Currently there are two forms of ECL ,ECL 10K and ECL 100K .Emitter coupled logic is

basically based on the used of current steering switch such a switch can be most conveniently using the differential pair.

DIODE TRANSISTOR LOGIC= mainly the diode transistor logic consists diodes and

transistors that comprise the DTL. It is the class of digital circuit is the direct ancestor

Is transistor transistor logic .it is so called because the logic gate performed by the diode

network and amplifying function is performed by a transistor .They refers the designing

and fabricating the circuits.

2 Write the type of digital IC packaging.

The digital packaging is the last stage of fabrication in the semiconductor .the common ic packaging is the through hole technology and surface mount technology ,chip carrier ,chip scale package and flat package they are all digital ic packaging.

3 Which Logic family is better to implement the integrated circuit?

The BICOMOS logic family has the better implement for integrated circuits .Because behind the reason is that it is the most fastest logic family than other logic families . The BICOMOS used to design in many circuits . These circuits easy to use and has simpler Components as well as less components .BICOMOS TTL is more used in the circuits.

Lab Task

- Design the circuits in MULTISIM according to circuit diagrams given above for all Logic IC's attach the screenshots of circuit, input sequence, input and output waveform and also show the truth tables.

Conclusion

I Achieve in this lab about logics gates and logics families , I learnt in this portion how to design different IC's on the multisim.

